

Null-Space Reinforcement Learning for Trajectory Optimization of Free-Floating Space Manipulators under Inertia Changes^{*}

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Abstract: This paper presents NS-TD3, a null-space reinforcement learning framework for trajectory planning of a 7-DOF free-floating space manipulator performing repetitive module transport under abrupt inertia changes. The one-dimensional kinematic redundancy is parameterized by a scalar α , and a TD3 agent learns the optimal $\alpha(t)$ policy that restores base attitude across full forward-and-homing cycles—a task beyond any instantaneous strategy when payload attachment shifts the inertia coupling matrix mid-cycle. Without explicit knowledge of the inertia-change event, NS-TD3 reduces terminal attitude error by 67% over the minimum-norm (MN) baseline and 42% over a receding-horizon QP (RH-QP) using the same inertia model with explicit mode switching, while satisfying joint constraints and maintaining end-effector accuracy at pseudoinverse-level computational cost.

Keywords: Free-Floating Manipulator; Null-Space Control; Reinforcement Learning; Redundancy Resolution; Trajectory Optimization; Space Robotics.

1. INTRODUCTION

Unlike ground-based systems, manipulators on compact space platforms operate without a fixed base, so joint motion directly induces significant base motion. Moreover, such a manipulator repeatedly performs module attachment, detachment, and satellite docking; each operation induces reaction forces on the platform base, and the resulting attitude disturbances accumulate over cyclic tasks, seriously compromising platform mission integrity and demanding specialized GNC techniques.

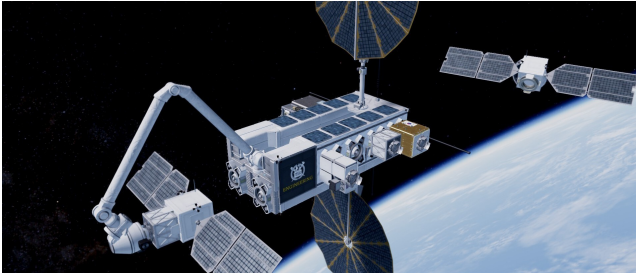


Fig. 1. Rendering of a compact space platform and manipulator.

To address these challenges, prior work has explored model-based planners, MPC, and adaptive controllers (Starek et al., 2012; Alizadeh and Zhu, 2024), though these approaches tend to be sensitive to model errors and struggle to adapt to abrupt inertia changes. Exploiting

kinematic redundancy offers a complementary direction: reaction null-space methods reduce instantaneous base disturbance by projecting self-motion into the null-space of the coupling matrix, but this projection does not guarantee end-effector tracking. Analytic approaches that greedily choose a null-space scalar to minimize instantaneous base recoil are locally optimal yet cannot resolve the cyclic asymmetry introduced when payload attachment shifts the inertia coupling mid-cycle, i.e., the same return trajectory produces a different base response after the shift, and no instantaneous policy can anticipate this.

In recent years, hybrid architectures integrating model-based control with reinforcement learning have proven effective for constrained robot control (Li et al., 2023; Testa et al., 2025). Among these, null-space-based redundancy resolution (Umetani and Yoshida, 1989) offers a particularly clean structure: the kinematic null-space projector analytically enforces end-effector constraints while the RL agent optimizes the remaining self-motion degree of freedom. Prior null-space RL work applied DDPG and SAC to self-motion on ground-fixed (Hua et al., 2021; Shen et al., 2022) and free-floating (Huang et al., 2023) robots, demonstrating joint limit handling and manipulability management. However, these studies target collision avoidance on ground-fixed or free-floating robots and do not address cyclic base attitude restoration under mid-cycle inertia changes that is the problem studied in this paper.

This paper proposes a null-space reinforcement learning framework in which the one-dimensional kinematic redundancy of a 7-DOF free-floating space manipulator is parameterized by a scalar α , and a TD3 agent directly

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learns the $\alpha(t)$ policy that minimizes residual base attitude disturbance across full forward-and-homing cycles under abrupt inertia changes caused by payload attachment. The contributions of this work are as follows: (i) A scalar null-space parameterization that reduces redundancy resolution to a single variable α , enabling direct policy learning in the one-dimensional null-space without high-dimensional action design. (ii) Quantitative demonstration that the learned TD3 policy reduces terminal base attitude error by 67% over the minimum-norm solution and 42% over a receding-horizon QP baseline that uses the same inertia model, while satisfying joint constraints and maintaining end-effector accuracy at pseudoinverse-level computational cost. (iii) The TD3 policy implicitly adapts to mid-cycle inertia changes, achieving superior attitude restoration without requiring event detection or explicit mode switching.

2. SYSTEM MODELING

2.1 Free-floating Dynamics

The space manipulator operates as a free-floating system without a fixed base. The equation of motion for a space robot with an n Degree-of-Freedom (DOF) manipulator is governed by

$$\begin{bmatrix} H_{bb} & H_{bm} \\ H_{bm}^T & H_{mm} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{x}}_b \\ \ddot{\mathbf{q}} \end{bmatrix} + \begin{bmatrix} \mathbf{c}_b(\mathbf{q}, \dot{\mathbf{q}}) \\ \mathbf{c}_m(\mathbf{q}, \dot{\mathbf{q}}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ \boldsymbol{\tau} \end{bmatrix} \quad (1)$$

where $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \in \mathbb{R}^n$ are the joint angles, velocities, and accelerations; $\dot{\mathbf{x}}_b, \ddot{\mathbf{x}}_b \in \mathbb{R}^6$ are the base velocity and acceleration; $H_{bb} \in \mathbb{R}^{6 \times 6}$, $H_{bm} \in \mathbb{R}^{6 \times n}$, $H_{mm} \in \mathbb{R}^{n \times n}$ are the base, coupling, and joint inertia submatrices; and $\mathbf{c}_b, \mathbf{c}_m$ are the Coriolis/centrifugal terms (Wilde et al., 2018). Since the base is unactuated, the control input reduces to joint torques $\boldsymbol{\tau} \in \mathbb{R}^n$ alone.

Under zero initial momentum, integrating the base row of (1) gives

$$H_{bb}\dot{\mathbf{x}}_b + H_{bm}\dot{\mathbf{q}} = \mathbf{0} \quad (2)$$

Solving (2) for $\dot{\mathbf{x}}_b$ gives the kinematic coupling

$$\dot{\mathbf{x}}_b = -H_{bb}^{-1}H_{bm}\dot{\mathbf{q}} \quad (3)$$

Equation (3) shows that base disturbance is proportional to H_{bb}^{-1} ; for compact platforms where $H_{bb} \gg H_{mm}$, this coupling is non-negligible and must be addressed explicitly.

As illustrated in Fig. 2, the manipulator extracts a module from a visiting satellite docked at the front port and transfers it to a side port (forward motion), after which the module is rigidly attached to the station structure. This attachment causes an abrupt increase in H_{bb} . As (3) shows, the base response is governed by $-H_{bb}^{-1}H_{bm}$; hence, even an identical return trajectory produces a different base motion after the inertia change, and residual drift accumulates over repeated operations.

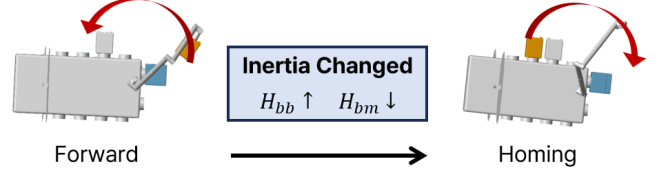


Fig. 2. Residual base attitude drift caused by abrupt inertia change upon module attachment.

2.2 Kinematics and Null-Space

Since the module targets are defined with respect to the station's base frame, the kinematics are described by the standard manipulator Jacobian $J_m(\mathbf{q}) \in \mathbb{R}^{6 \times 7}$,

$$\dot{\mathbf{x}}_{ee} = J_m(\mathbf{q})\dot{\mathbf{q}} \quad (4)$$

With $\text{rank}(J_m) = 6$ for a 7-DOF arm tracking a 6D task, $\dim \mathcal{N}(J_m) = 7 - 6 = 1$, so the arm has exactly one internal self-motion degree of freedom that leaves the end-effector unaffected. The null-space basis vector is extracted via SVD,

$$J_m = U\Sigma V^T, \quad \mathbf{v}_{\text{null}} \triangleq V(:, 7), \quad J_m \mathbf{v}_{\text{null}} = \mathbf{0} \quad (5)$$

The complete family of joint-velocity solutions to (4) is

$$\dot{\mathbf{q}} = J_m^\dagger \dot{\mathbf{x}}_{des} + (I - J_m^\dagger J_m) \mathbf{z}, \quad \mathbf{z} \in \mathbb{R}^7 \quad (6)$$

where $J_m^\dagger$ is the Moore–Penrose pseudoinverse and $(I - J_m^\dagger J_m)$ is the null-space projector. Since $\dim \mathcal{N}(J_m) = 1$, every self-motion is a scalar multiple of $\mathbf{v}_{\text{null}}$, and (6) reduces to

$$\dot{\mathbf{q}} = \underbrace{J_m^\dagger \dot{\mathbf{x}}_{des}}_{\dot{\mathbf{q}}_p} + \alpha \mathbf{v}_{\text{null}}, \quad \alpha \in \mathbb{R} \quad (7)$$

where $\dot{\mathbf{q}}_p$ is the minimum-norm particular solution ($\alpha = 0$). As illustrated in Fig. 3, the solutions form a line in $\dot{\mathbf{q}}$ -space, with $\alpha = 0$ at the minimum-norm point.

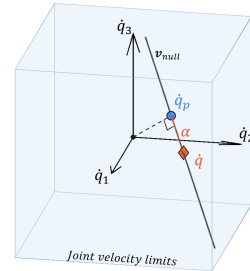


Fig. 3. Null-space geometry: the solution line in joint velocity space.

3. NULL-SPACE REINFORCEMENT LEARNING FRAMEWORK

The proposed framework, illustrated in Fig. 4, maps the one-dimensional kinematic redundancy of the 7-DOF manipulator to a scalar policy learning problem. At each low-level control step, a pseudoinverse-based resolver computes

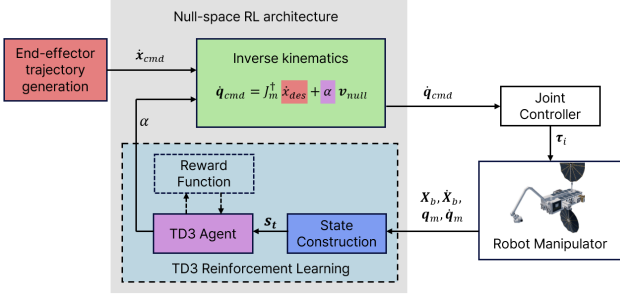


Fig. 4. Architecture of the proposed null-space reinforcement learning framework.

the joint velocity command; every $T_s = 2$ s, a TD3 agent outputs a rate command $\dot{\alpha} \in [-1, 1]$, which is integrated to update the null-space coefficient α for the next period.

3.1 Trajectory Generation

The motion profile for each phase (forward transport and homing) is pre-planned as a fifth-order polynomial in end-effector Cartesian space. Given boundary conditions on position, velocity, and acceleration at the segment endpoints, the polynomial uniquely determines smooth $\mathbf{x}_{cmd}(t)$ and $\dot{\mathbf{x}}_{cmd}(t)$ profiles with zero velocity at both endpoints.

At each control step, the desired end-effector velocity incorporating feedback is computed via Closed-Loop Inverse Kinematics (CLIK)

$$\dot{\mathbf{x}}_{des} = \dot{\mathbf{x}}_{cmd} + K_{CLIK}(\mathbf{x}_{cmd} - \mathbf{x}_{curr}) \quad (8)$$

3.2 Null-Space RL Parameterization

The Jacobian $\mathbf{J}_m$ is decomposed via SVD following (5). To prevent velocity spikes near kinematic singularities, the pseudoinverse is formed with Damped Least Squares (DLS) regularization. Denoting the singular values as σ_i , the damped inverse singular values are

$$\tilde{\sigma}_i = \frac{\sigma_i}{\sigma_i^2 + \lambda_i^2}, \quad \lambda_i = \begin{cases} \lambda_{\max} \left(1 - \left(\frac{\sigma_i}{\sigma_{th}} \right)^2 \right) & \sigma_i < \sigma_{th} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

giving $\mathbf{J}_m^\dagger = \mathbf{V} \text{diag}(\tilde{\sigma}_i) \mathbf{U}^T$, the particular solution $\dot{\mathbf{q}}_p = \mathbf{J}_m^\dagger \dot{\mathbf{x}}_{des}$, and the null-space basis $\mathbf{v}_{null} = \mathbf{V}(:, 7)$.

The one-dimensional null-space from (7) is parameterized by the scalar coefficient $\alpha(t)$:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_p + \alpha \mathbf{v}_{null} \quad (10)$$

Setting $\alpha = 0$ recovers the minimum-norm particular solution $\dot{\mathbf{q}}_p$ with no self-motion. Rather than directly setting α , the TD3 agent outputs a normalized rate command $\dot{\alpha} \in [-1, 1]$; α is updated at each RL decision by

$$\alpha_{k+1} = \text{clip}(\alpha_k + \alpha_{\max} \frac{T_s}{2} \dot{\alpha}_k, -\alpha_{\max}, \alpha_{\max}) \quad (11)$$

where $\alpha_{\max} = 0.1$ is chosen heuristically to keep self-motion within a physically admissible range.

3.3 Joint Constraint Enforcement

Prior to the velocity clamp, α is soft-damped when Joint 2 enters a warning zone of width d_{warn} , but *only if* the self-motion is pushing toward the limit:

$$\alpha \leftarrow \begin{cases} s(q_2) \alpha, & \dot{\alpha} v_{n,2} > 0 \\ \alpha, & \text{otherwise} \end{cases}, \quad s(q_2) = \left(\frac{d_\ell}{d_{warn}} \right)^2 \quad (12)$$

where d_ℓ is the remaining distance to the active position limit, $v_{n,2}$ is the second element of $\mathbf{v}_{null}$, and the product $\dot{\alpha} v_{n,2} > 0$ indicates that the RL command is driving Joint 2 toward the limit. Self-motion in the corrective direction is unaffected; the quadratic scale s reduces only the limit-approaching null-space component, leaving $\dot{\mathbf{q}}_p$ intact and suppressing chattering well before the hard boundary. The damped candidate is then element-wise clamped to velocity limits with bounds linearly tapered near position limits:

$$\dot{\mathbf{q}}^* = \text{clip}(\dot{\mathbf{q}}, \mathbf{l}(\mathbf{q}), \mathbf{u}(\mathbf{q})) \quad (13)$$

3.4 TD3-based MDP Formulation

The null-space control problem is formulated as a Markov Decision Process (MDP) and solved with the Twin Delayed Deep Deterministic Policy Gradient (TD3) (Fujimoto et al., 2018). TD3's Clipped Double Q-learning and Delayed Policy Updates prevent the Q-value overestimation that destabilizes DDPG (Lillicrap et al., 2016) in the nonlinear free-floating environment.

1) *Observation Space*: The observation vector $\mathbf{s}_t \in \mathbb{R}^{30}$ is assembled and normalized as

$$\mathbf{s}_t = \left[\frac{\mathbf{e}_{ee}}{s_{ee}}, \mathbf{q}_b, \frac{\boldsymbol{\omega}_b}{s_\omega}, \frac{\mathbf{v}_b}{s_v}, \frac{\mathbf{q}_{arm}}{s_q}, \frac{\dot{\mathbf{q}}_{arm}}{s_{\dot{q}}}, \alpha_{t-1}, d, \frac{t}{T_f} \right]^T \quad (14)$$

comprising EE position error, base quaternion, base angular and linear velocities, joint angles and velocities, previous null-space coefficient, docked flag, and normalized time; each divided by its expected maximum value.

2) *Action Space*: The agent outputs a rate command $\dot{\alpha} \in [-1, 1]$, held constant for $T_s = 2$ s. The null-space coefficient α is updated by Euler integration and clamped to $[-\alpha_{\max}, \alpha_{\max}]$, producing smooth continuous self-motion. Feasibility is guaranteed by the box clamp (13) regardless of the agent's output.

3) *Reward Function*:

$$r_t = (r_{track} + r_{base} + r_{smooth} + r_{limit} + r_{alive}) \quad (15)$$

r_{track} is a Gaussian on end-effector position error, with its weight reduced after module docking to shift emphasis toward attitude restoration. The attitude term uses a time-varying weight that increases quadratically over the homing phase

$$r_{base} = w(t) \left(1 - \frac{\theta_{err}}{\theta_{ref}} \right)$$

$$w(t) = w_{min} + (w_{max} - w_{min}) \left[\frac{t - t_1}{T_f - t_1} \right]_+^2 \quad (16)$$

where $\theta_{err} = 2 \arccos |q_{b,w}|$. This directs the agent to prioritize tracking early and attitude restoration late. At the final step ($t \geq T_f - T_s$), a one-time terminal bonus $r_{terminal} = w_{term} \left(1 - \frac{\theta_{err}}{\theta_{ref}/2} \right)$ is added to r_{base} to sharply reward near-zero terminal attitude error. r_{smooth} penalizes abrupt action changes to encourage smooth self-motion.

r_{limit} closes a gap in (13): the clamp prevents per-step violations but cannot discourage persistent limit-directed self-motion. When Joint 2 enters a margin zone near a position limit,

$$r_{limit} = -w_l \rho [\sigma_\ell \dot{\alpha}]_+^2 \quad (17)$$

where $\rho \in [0, 1]$ is the normalized proximity to the active limit, $\sigma_\ell = +1$ near the lower limit and -1 near the upper limit, and $[\cdot]_+ = \max(0, \cdot)$ selects only the limit-approaching component. Actions directed away from the boundary incur no penalty, preserving the agent's ability to self-correct. The survival bonus $r_{alive} = 1.0$ provides a constant positive baseline.

The training configurations are summarized in Table 1.

Table 1. Hyperparameters of the TD3 Agent

Parameter	Value
Learning Rate (Actor / Critic)	$1 \times 10^{-3} / 2 \times 10^{-3}$
Discount Factor (γ)	0.996
Batch Size	512
Replay Buffer Size	1×10^5
Target Smooth Factor (τ)	5×10^{-3}
Exploration Noise (σ)	0.4 w/ decay, min 0.05
Policy Noise (σ)	0.1
N-step Return	10
Action Update Interval (T_s)	2 s
Max Episodes / Steps	8000 / 105

4. SIMULATION RESULTS

Two simulation environments validate the framework in increasing order of complexity, with MN ($\alpha = 0$) and RH-QP serving as comparison baselines in Section 4.2.

4.1 3-DOF Planar Arm

A 3-DOF planar free-floating arm executing a rectangular end-effector trajectory via the Generalized Jacobian Matrix yields a one-dimensional null-space whose structure can be visualized directly in 3-D joint velocity space. System parameters are listed in Table 2. Since joints 1 and 2 primarily determine end-effector positioning, the null-space direction aligns primarily with $\dot{q}_3$, as visualized in Fig. 6; accordingly, the RL policy exploits joint 3 significantly more than the MN baseline, shown in Fig. 5. Fig. 7 and Table 3 compare the resulting base attitude error, confirming that the RL policy achieves lower terminal error than MN.

Table 2. 3-DOF planar arm parameters

	Dimension [m]	Mass [kg]
Base	2×1	500
Link 1	1.05	10
Link 2	1.05	10
Link 3	0.3	5

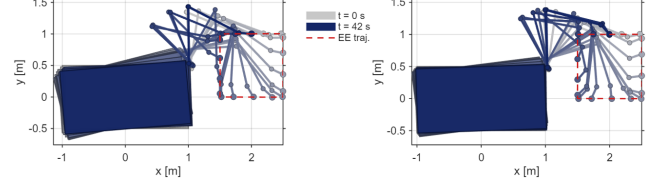


Fig. 5. Simulation snapshots of the planar 3-DOF arm: MN baseline (left) and RL policy (right).

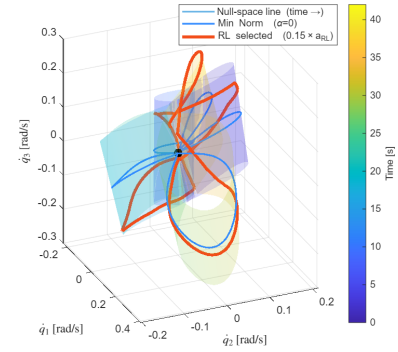


Fig. 6. Null-space direction in joint velocity space and $\alpha(t)$ trajectory (3-DOF planar arm).

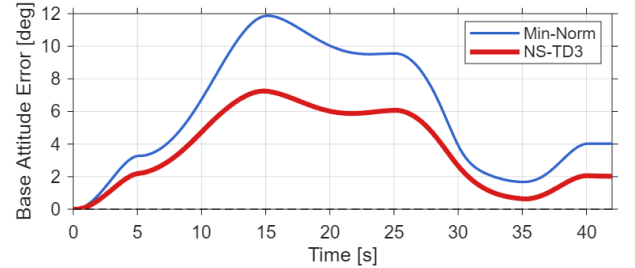


Fig. 7. Base attitude error $\theta_{err}(t)$ over time: MN baseline vs. RL policy (3-DOF).

Table 3. Base attitude error comparison

	MN	RL
Max error [deg]	11.8315	7.2394
Final attitude error [deg]	4.02367	2.02208

4.2 7-DOF Free-Floating Manipulator

The full framework is validated on a 7-DOF free-floating manipulator with total arm length 4.2m, modeled after Canadarm2 in a MATLAB/Simscap Multibody environment, equipped with a PD torque controller. The manipulator executes forward and homing motions under the module-attachment inertia change described in Section 2. Module attachment is modeled by rigidly coupling the payload to the station, triggering the abrupt inertia change. Link inertial properties are listed in Table 4 and simulation parameters in Table 5.

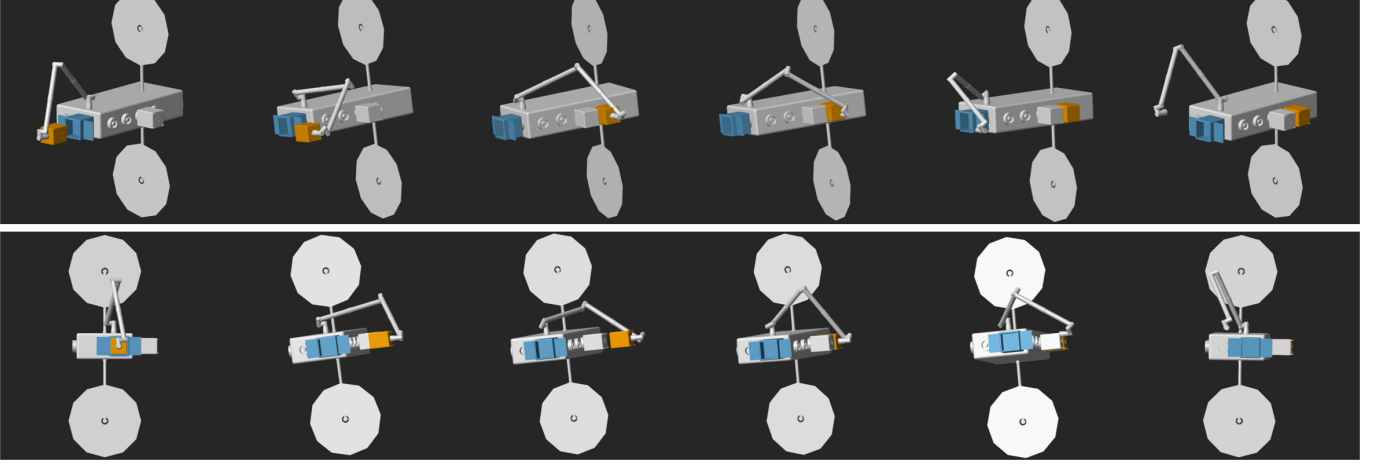


Fig. 8. Simulation snapshots of the 7-DOF free-floating manipulator: 3D perspective view (top) and front view (bottom).

Table 4. Link inertial properties

Link	Mass [kg]	I_{xx}	I_{yy}	I_{zz}
		[kg·m ²]		
Station	3000	776.9	2512.8	2739.1
Link 1	10	0.0940	0.0925	0.0240
Link 2	5	0.0133	0.0118	0.0110
Link 3	70	0.1352	18.787	18.781
Link 4	70	0.1352	18.787	18.781
Link 5	5	0.0133	0.0110	0.0118
Link 6	5	0.0133	0.0118	0.0110
Link 7	15	0.0512	0.0518	0.0389

Table 5. Simulation parameters (7-DOF)

Parameter	Value
Module Mass	40 kg
Joint Velocity Limit	± 10 deg/s
Joint 2 Angle Limit	-30° to -135°
Torque Limit	± 3.0 Nm
Simulation Time Step	0.005 s
Inverse Kinematics Time Step	0.1 s

Three policies are compared: (i) **MN** (minimum-norm, $\alpha = 0$), which produces no self-motion; (ii) **RH-QP**, a receding-horizon QP that optimizes α over a $H = 10$ -step window, switching objectives between phases— $\min_{\alpha} \|B(\dot{\mathbf{q}}_p + \alpha \mathbf{v}_{\text{null}})\|^2$ during forward motion and $\min_{\alpha} \|B_{\text{ang}}(\dot{\mathbf{q}}_p + \alpha \mathbf{v}_{\text{null}}) - k_p e_{\text{att}}\|^2$ during homing—using the same inertia model available to the proposed method; and (iii) the **NS-TD3** policy, which learns $\alpha(t)$ from full-cycle experience without explicit mode switching.

Fig. 8 illustrates the learned elbow behavior in simulation. The NS-TD3 policy exploits the null-space degree of freedom—physically the elbow rotation governed primarily by Joint 2—to actively reposition the arm throughout the cycle. Because this self-motion does not affect end-effector tracking, it is transparent to task execution; yet the resulting elbow posture difference (visible in both the 3D and front views) is exactly the mechanism that restores base attitude under inertia changes.

Fig. 9 shows the evolution of $\alpha(t)$ for each policy (top) and the corresponding Joint 2 trajectory (bottom), where the largest configuration difference among the three policies is observed. All three satisfy the joint angle con-

straints throughout the cycle. Notably, NS-TD3 produces a markedly different $\alpha(t)$ trajectory from RH-QP, indicating that the learned policy has found a globally distinct self-motion strategy rather than a local variant of the receding-horizon solution.

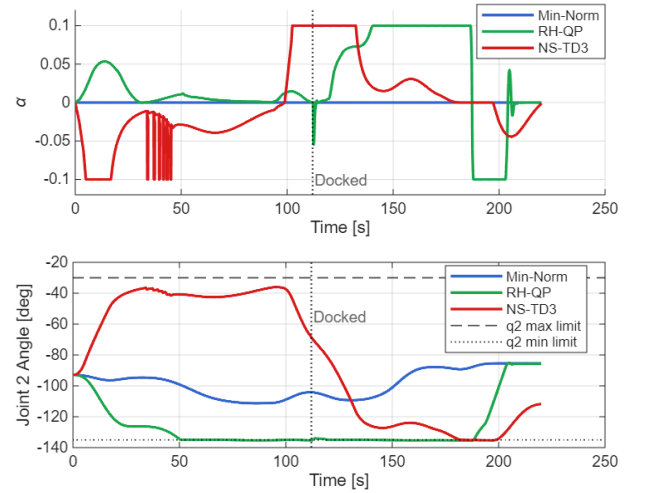


Fig. 9. $\alpha(t)$ for MN, RH-QP, and NS-TD3 (Top). Joint 2 angle trajectory (Bottom).

Fig. 10 compares base attitude error $\theta_{\text{err}}(t)$ over the full cycle: NS-TD3 achieves the lowest terminal error, confirming that cycle-level optimization outperforms both the instantaneous MN baseline and the short-horizon RH-QP strategy under mid-cycle inertia changes. Per-step computation times for the three policies are compared in Fig. 11 (all timing measurements were performed on an AMD Ryzen 7 7800X3D), and quantitative results are summarized in Table 6.

Table 6. Performance comparison (7-DOF free-floating manipulator)

Metric	MN	RH-QP	NS-TD3
Terminal att. error [deg]	7.32	4.24	2.45
Mean EE error [cm]	0.29	0.62	0.33
Mean comp. time [ms]	0.45	2.78	0.52

The proposed NS-TD3 reduces terminal base attitude error by 67% relative to MN and by 42% relative to

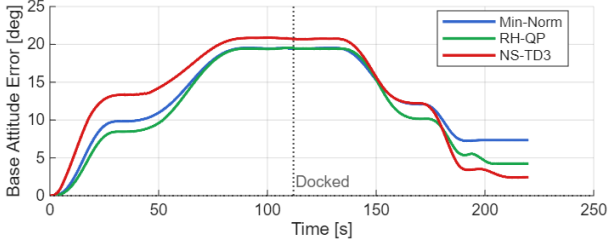


Fig. 10. Base attitude error $\theta_{err}(t)$ for MN, RH-QP, and NS-TD3.

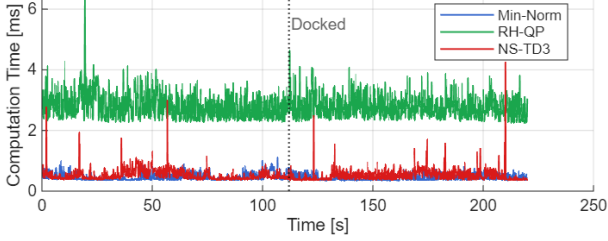


Fig. 11. Per-step computation time for MN, RH-QP, and NS-TD3.

RH-QP, while maintaining end-effector tracking accuracy comparable to MN (0.33 cm vs. 0.29 cm). RH-QP improves over MN by exploiting a finite prediction horizon, yet its mode-switched, short-horizon objective cannot compensate for the asymmetric inertia shift accumulated over the full forward-and-homing cycle; the TD3 policy, trained on complete cycles, encodes this global compensation implicitly. Crucially, the mean computation time of NS-TD3 (0.52 ms) is less than one-fifth that of RH-QP (2.78 ms) and adds only 0.07 ms over the minimum-norm pseudoinverse (0.45 ms), confirming that cycle-level attitude restoration is achieved with negligible additional computational overhead.

5. CONCLUSION

This paper presented a null-space reinforcement learning framework in which a TD3 agent learns the optimal scalar self-motion parameter $\alpha(t)$ for a 7-DOF free-floating space manipulator performing repetitive module transport under abrupt inertia changes. By operating entirely within the one-dimensional null-space, the policy improves terminal base attitude stability globally across full forward-and-homing cycles without degrading end-effector tracking accuracy. Simulation results show that NS-TD3 significantly outperforms both the MN baseline and the model-aware RH-QP, while satisfying joint constraints at pseudoinverse-level computational cost. Notably, the learned $\alpha(t)$ trajectory diverges markedly from the RH-QP solution, indicating that the TD3 policy has discovered a globally distinct self-motion strategy that the short-horizon optimizer cannot reach—a key indicator that cycle-level RL optimization offers a qualitatively different solution beyond local improvement.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used ChatGPT in order to correct grammatical errors. After

using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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